A Census of Multi-Colour Langton's Ants up to Twelve Colours, with an Exact Escape Formula and a Highway Proof for the Family $\mathtt{L}^{K-1}\mathtt{R}$

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Abstract

A multi-colour Langton's ant is given by a rule string of K letters `L` and `R`. We run every rule string with $K = 2$ to 12 colours for one million steps and sort the 4,015 distinct machines that remain, after removing mirror images and repeated strings, by the shape they grow: 2,723 chaotic, 790 square, 416 highway, 83 symmetric and 3 triangle. Every symmetric machine has a rule whose runs of equal letters, read in a circle, all have even length. We show that 164 of the highway machines never read the last letters of their rule and walk the same path as another machine, leaving 252 different highway paths, and that highways starting with `R` in second place do not occur for fewer than seven colours. Our main result concerns the family $A_K = \mathtt{L}^{K-1}\mathtt{R}$. We prove that, for every $K \geq 3$, A_K makes exactly 7,488 right turns, at the same squares and in the same directions for every K, before it first leaves the square box of side 2,998 around its start (the edge, Section 2), which happens at step $26{,}969 + 11{,}980\,(K-3)$, and that it then builds a diagonal highway forever, advancing two squares diagonally every $16K - 12$ steps. The proof is a single computer run in which K is kept as a symbol: every square's colour is stored as an affine expression $a + bK$, and every spin of the ant round a 2×2 loop is jumped over in one step. All code and data are public.

Keywords: Langton's ant, turmites, cellular automata, highway, symbolic simulation, computer-assisted proof.
MSC 2020: 37B15, 68Q80.

1 Introduction

Langton's ant [1] is a single head moving on an infinite square grid of white squares. On a white square it turns right, on a black square it turns left; it then flips the colour of the square and steps forward. Starting from an all-white grid, it wanders chaotically for about 10,000 steps and then builds a "highway": a pattern that repeats every 104 steps and moves diagonally away forever. That the ant always escapes to infinity is a theorem [4]; that it always ends on a highway, from any finite starting pattern, is still open. The ant is also capable of universal computation [5], so questions about its long-run behaviour are hard in general.

Turk and Propp generalised the ant to K colours [2, 3]. A rule string $w = w_0 w_1 \cdots w_{K-1}$ of letters `L` and `R` says which way to turn on each colour; the square's colour then advances by one, modulo K. The classic ant is `RL`. Gale, Propp, Sutherland and Troubetzkoy [3] studied these ants and showed that rules built from repeated pairs of letters give symmetric patterns. Many individual rules are known for their pictures: `LLRR` grows a symmetric shape, `LRRRRRLLR` fills a square, and `RRLLLRLLLRRR` grows a triangle.

We were not able to find a complete table of how often each kind of shape occurs for each number of colours. This paper gives one for $K \leq 12$ (Section 3), reports patterns in the rule

strings that predict the shape (Section 4), and then proves an exact result for the infinite family $A_K = \mathtt{L}^{K-1}\mathtt{R}$ (Sections 5 and 6). The proof method, running the ant once with K as a symbol, may apply to other families. Throughout, the census results are *experimental*: they describe each machine after a fixed number of steps and depend on our classification tests. Theorem 8 and the lemmas of Sections 4.1 and 6 are *proved*.

2 Definitions

Definition 1 (Ant)**.** A *rule* is a string $w = w_0 \cdots w_{K-1} \in \{\mathtt{L}, \mathtt{R}\}^K$, $K \geq 2$. A *state* is a position $p \in \mathbb{Z}^2$, a direction $d \in \{0, 1, 2, 3\}$ (up, right, down, left) and a colouring $\sigma : \mathbb{Z}^2 \to \{0, \ldots, K-1\}$ that is 0 on all but finitely many squares. One *step* reads $c = \sigma(p)$, turns right ($d \leftarrow d+1 \bmod 4$) if $w_c = \mathtt{R}$ and left ($d \leftarrow d-1 \bmod 4$) if $w_c = \mathtt{L}$, sets $\sigma(p) \leftarrow c+1 \bmod K$, and moves p one square in direction d. We use screen coordinates: up is $(0, -1)$, right is $(1, 0)$. The ant starts at $p = (0, 0)$, facing up, with $\sigma \equiv 0$.

Since every visit adds one to the colour, $\sigma(p)$ is the number of visits to p so far, modulo K. On its n-th visit to a square the ant reads letter $w_{(n-1) \bmod K}$.

Definition 2 (Reaching the edge)**.** Let $B = \{(x, y) : -1499 \leq x \leq 1498,\ -1499 \leq y \leq 1498\}$. The ant *reaches the edge at step t* if the move made in step t (steps counted from 0) is the first to take it outside B. This is the same as running on a 3000×3000 grid started at its centre and stopping one square from its border; all step counts in this paper use it.

Machines that are the same. Two rules can give the same ant:

- *Mirror twins.* Swapping every L and R gives the mirror image of the same path. We keep one rule of each pair, the one that comes first alphabetically (it starts with L).
- *Repeated strings.* If $w = u^j$ for a shorter string u (for example $\mathtt{RLRL} = (\mathtt{RL})^2$), the turn on a square depends on the number of visits modulo $|u|$ only, so w walks exactly the path of u. We keep only *primitive* rules, which are not powers of a shorter string.

After both reductions there are $1, 3, 6, 15, 27, 63, 120, 252, 495, 1{,}023$ and 2,010 machines for $K = 2, \ldots, 12$, which is 4,015 in total. (This is half the number of primitive binary words of length K.)

3 The census for $K \leq 12$

Method. All 8,188 rule strings with $2 \leq K \leq 12$ were run for 10^6 steps on a 3000×3000 grid. Each final picture was given one class by the tests below, applied in this order to the bounding box of the visited squares. The thresholds were tuned by looking at random samples of every class.

Class	Test on the picture after 10^6 steps
Highway	The ant reached the edge, and under 5% of the box it crossed is coloured.
Triangle	Width/height ratio over 0.8; one half of the box, cut along a diagonal, is under 3% coloured and the opposite half over 30%.
Symmetric	The colours agree with the picture's 180° turn or one of its mirror images on over 90% of coloured squares, and under 95% of the box is coloured.
Square	Width/height ratio over 0.9, and over 86% of the box coloured (or over 80% with mostly coloured borders).
Chaotic	Everything else.

Table 1: Distinct machines by number of colours K and class, after 10^6 steps.

K	Highway	Square	Triangle	Symmetric	Chaotic	Machines
2	1	0	0	0	0	1
3	1	0	0	0	2	3
4	2	0	0	2	2	6
5	1	2	0	0	12	15
6	2	3	0	5	17	27
7	9	12	0	0	42	63
8	13	22	0	10	75	120
9	24	47	0	0	181	252
10	54	92	0	25	324	495
11	107	213	1	0	702	1,023
12	202	399	2	41	1,366	2,010
All	416	790	3	83	2,723	4,015

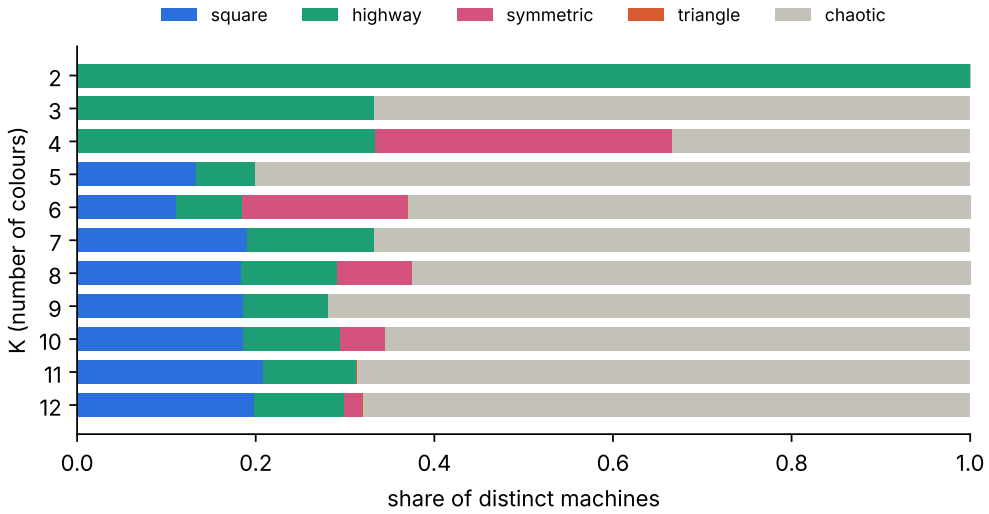


Figure 1: Share of each class among the distinct machines with K colours.

Results. Table 1 gives the counts and Figure 1 the shares; Figure 2 shows two machines of each class.

About two machines in three grow a chaotic blob. From $K = 7$ on, about one in five grows a square, roughly twice as many as build a highway. Only three triangles appear: `LLRLRRLLLLL` ($K = 11$) and `LLLLLLRRLRLL` and `LLRRRLRRRLLL` ($K = 12$); the last is the mirror image of the known triangle rule `RRLLLRLLLRRR`.

4 Patterns in the rule strings

Observation 3 (Even runs and symmetry)**.** Read a rule in a circle and split it into runs of equal letters. All 83 symmetric machines have every run of even length (for example `LLRR`, `LRRL`, `LLLRRL`). Of the 104 machines with this property, the other 21 were classed square (16) or chaotic (5); the 16 squares are themselves highly symmetric in colour. In particular no symmetric machine exists for odd K, since an odd length cannot be split into even runs.

This matches the result of [3], which shows that such rules give patterns that are symmetric

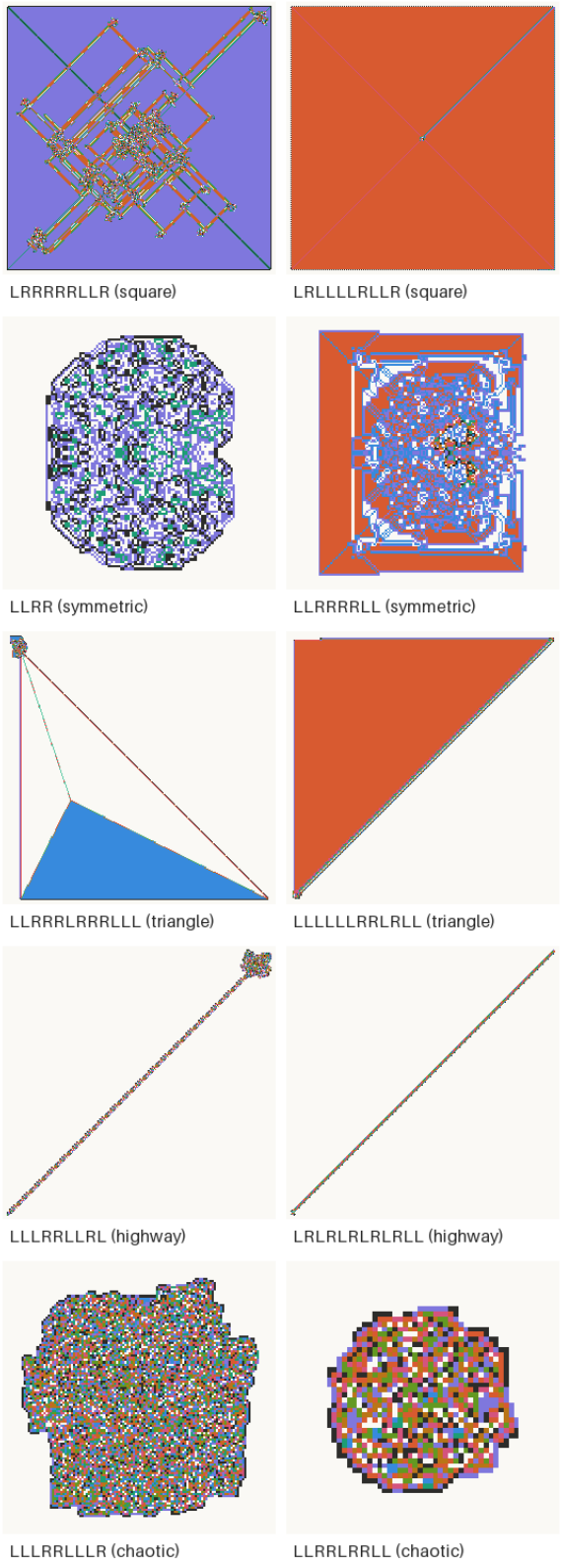


Figure 2: Two machines of each class, cropped to the coloured area (white is colour 0), after 10^6 steps; highways are shown when they reach the edge.

whenever the ant returns to its starting square, not at every step; the 5 "chaotic" cases are most likely pictures taken between two such moments.

Observation 4 (The first letters)**.** For $6 \le K \le 12$, the first three letters lean strongly towards a class:

Rule starts with	Machines	Square	Highway	Chaotic
LRR	998	39%	0%	59%
LRL	997	18%	7%	75%
LLR	998	16%	20%	62%
LLL	997	6%	14%	76%

Observation 5 (Highways need two equal first letters, until $K = 7$)**.** For $K = 3$ to 6, every highway rule starts with LL: the ant turns the same way on a square's first two visits. The first highway starting with LR appears at $K = 7$ (LRLLRRL, which reaches the edge only at step 314,536). The numbers of highways starting LL and LR for $K = 3, \ldots, 12$ are 1/0, 2/0, 1/0, 2/0, 8/1, 11/2, 21/3, 43/11, 89/18 and 163/39. Over all K, rules starting LL build highways 17.0% of the time and rules starting LR 3.7% of the time; without the copies of Section 4.1 the first rate is 8.8%.

Observation 6 (Slow growers)**.** 228 machines that are not highways are still under 40 squares wide after 10^6 steps. For example LLLRR is 28 squares wide after 10^6 steps, 36 after 1.6×10^7 and 49 after 1.024×10^9, gaining about four squares each time the number of steps is multiplied by four; its picture is a slowly growing disc.

4.1 Hidden copies among the highways

The step at which a highway machine reaches the edge revealed that many of them are the same machine. For instance every highway rule beginning with LLRLLR reaches the edge at exactly step 26,969, whatever its length and later letters.

Lemma 7 (Unread letters)**.** *Suppose that, up to step T, the ant with rule w never visits any square more than M times, with $M < |w|$. Then up to step T it reads only $w_0, \ldots, w_{M-1}$, and every rule that starts with $w_0 \cdots w_{M-1}$ and has at least M letters walks the same path up to step T.*

Proof. On its n-th visit to a square the ant reads $w_{(n-1) \bmod |w|}$, and $n \le M < |w|$, so the index is $n - 1 \le M - 1$. A rule w' with the same first M letters gives the same turn at every step, by induction on the step. □

Measuring the largest number of visits to one square up to the edge, 164 of the 416 highway machines never read their last letters. They fall into four families (Table 2). In the first three, the letters read are a shorter rule written twice, so by the repeated-strings remark of Section 2 the path is that of the shorter ant. Grouping machines that read the same letters, there are 252 different highway paths among the 416 highway machines.

Table 2: Highway machines that never read their last letters before reaching the edge.

Letters read	Same path as	Most visits	Reaches edge at	Copies
LLRLLR	LLR ($K = 3$)	6	26,969	124
LLLRLLLR	LLLR ($K = 4$)	8	38,949	29
LLLLRLLLLR	LLLLR ($K = 5$)	10	50,929	6
LLLRRRLLLR	itself ($K = 10$)	10	56,928	5

5 The family $\mathtt{L}^{K-1}\mathtt{R}$

The first three families of Table 2 come from the rules $A_K = \mathtt{L}^{K-1}\mathtt{R}$, which turn left on every visit to a square except each K-th. Their edge steps are 26,969, 38,949 and 50,929 for $K = 3, 4, 5$: an arithmetic progression with difference 11,980. Direct runs confirm the progression for every K from 3 to 25 and for $K = 30, 40, 50, 75, 100, 150, 200, 300, 500$ and 1,000 (Figure 3). In every one of these runs the most visits to one square is exactly $2K$.

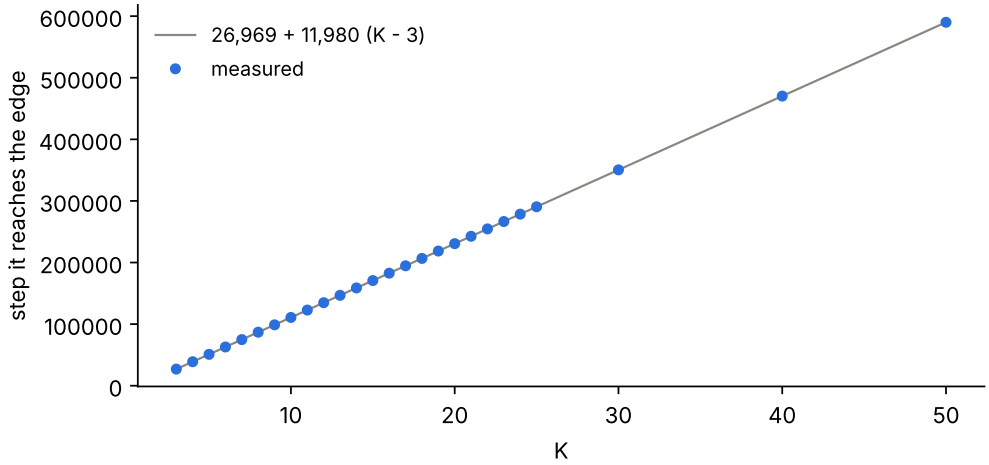


Figure 3: Step at which $\mathtt{L}^{K-1}\mathtt{R}$ reaches the edge, measured, against $26{,}969 + 11{,}980\,(K-3)$.

Theorem 8. *For every $K \geq 3$, the ant $A_K = \mathtt{L}^{K-1}\mathtt{R}$ started on an empty grid:*

(i) makes exactly 7,488 right turns before it reaches the edge, and the n-th right turn happens on the same square, leaving in the same direction, for every K;

(ii) reaches the edge at step $26{,}969 + 11{,}980\,(K-3) = 11{,}980\,K - 8{,}971$;

(iii) after its 7,600-th right turn, repeats forever a period of 10 right turns that takes $16K - 12$ steps and moves the ant by $(-2, 2)$, so it builds a diagonal highway and moves away to infinity.

Why it should be true. Runs for $K \in \{3, 4, 5, 6, 7, 10, 20\}$ show the mechanism. Between two right turns the ant makes only left turns, and four left turns in a row take it round a 2×2 loop and back to the same square in the same direction. We call this a *spin*: it adds one visit to each of four squares per lap, until one of them reaches its letter R. Every gap of left turns between two consecutive right turns has the form $g_3 + \gamma\,(K-3)$ with $\gamma \in \{0, 4\}$; exactly 2,995 gaps have $\gamma = 4$, and $2{,}995 \times 4 = 11{,}980$. Each extra colour adds one lap to those spins and changes nothing else. Likewise, on every square the number of visits is the $K = 3$ number plus $m\,(K-3)$ with $m \in \{0, 1, 2\}$: whole extra cycles, which leave every square the same number of visits away from its next R. The proof turns these observations into a statement for all K at once.

6 The proof

The proof is a computer run, `proof/symbolic.py` and `proof/highway.py` in the repository, that follows the ant for all $K \geq 4$ simultaneously. The case $K = 3$ is a single ordinary run (`proof/highway_concrete.py`, Section 6.6).

6.1 Symbolic colours

Fix $K \geq 4$. Every number the run needs is an *affine expression* $a + bK$ with integers a, b. We store the colour $\sigma(p) \in \{0, \ldots, K-1\}$ of each square as such an expression in one of two forms: $a + 0K$ with $a \geq 0$, or $a + 1K$ with $a \leq -1$. Over the whole run the constants stay small: $0 \leq a \leq 2$ in the first form and $-3 \leq a \leq -1$ in the second, so each expression is a genuine colour in $\{0, \ldots, K-1\}$ for every $K \geq 4$.

The run makes two kinds of decision.

1. *Is the ant on its* `R`*?* For A_K the ant turns right exactly when $\sigma(p) = K - 1$. The run answers *yes* exactly when the stored expression is $-1+1K$. This is correct for every $K \geq 4$: an expression $a + 0K$ with $a \leq 2$ never equals $K - 1$, and $a + 1K$ equals $K - 1$ only if $a = -1$.
2. *Comparisons.* Whether $e = a + bK$ is positive, negative or zero for all large K is decided by the sign of b (or of a when $b = 0$). Each such answer holds from some K on; the run records the largest such threshold. For this run it is $K_0 = 4$.

Equal expressions always give equal numbers, so whenever the run finds two colours equal they are equal for every $K \geq K_0$.

6.2 Jumping over spins

Lemma 9 (Spin)**.** *Let the ant be at square c_0 facing d, and let c_0, c_1, c_2, c_3 be the squares it visits if it turns left four times (so its fourth move returns it to c_0 facing d). Let $r_i = (K-1) - \sigma(c_i)$ be the number of visits c_i can still take while reading* `L`*, and suppose $n = \min_i r_i \geq 1$. Then the next $4n$ steps are n laps of left turns round c_0, c_1, c_2, c_3; afterwards the ant is again at c_0 facing d, each $\sigma(c_i)$ has grown by n, and no colour has wrapped round.*

Proof. In lap $\ell \in \{1, \ldots, n\}$ square c_i is visited with colour $\sigma(c_i) + \ell - 1 \leq \sigma(c_i) + n - 1 \leq \sigma(c_i) + r_i - 1 = K - 2$, so the ant reads `L` there and turns left. Four left turns bring it back to c_0 facing d. After n laps each colour is $\sigma(c_i) + n \leq K - 1$. □

The run applies Lemma 9 whenever all four r_i are positive, with n and the step count $4n$ kept as affine expressions; n is the least of the r_i, found with recorded comparisons. Otherwise it makes one ordinary step. Positions and directions stay ordinary integers throughout, because the order of decisions does not depend on K. Near the edge of B spins are not jumped, so the step at which the ant leaves B is found exactly.

6.3 Parts (i) and (ii)

The symbolic run reaches the edge after 7,488 right turns and 3,743 jumped spins, with step count $11{,}980\,K - 8{,}970$ moves, so the edge is reached at step $11{,}980\,K - 8{,}971$. Every decision on the way holds for all $K \geq 4$. Since positions and directions in the run are integers that do not involve K, the right turns are the same squares and directions for every $K \geq 4$. This proves (i) and (ii) for $K \geq 4$.

6.4 Part (iii): the highway goes on forever

For the t-th right turn, let p_t be the square where it happens, d_t the direction the ant faces after turning, S_t the colouring just after that square's colour is updated, and H_t the set of squares visited so far. The ant's motion after a right turn is determined by p_t, d_t and S_t. Fix right turns i and $j = i + P$, let V be the set of squares visited after right turn i, up to and including right turn j, let $D = p_j - p_i$, and let $f(c) = c \cdot D$ (dot product). Put $A = (V + D) \setminus V$.

Lemma 10 (Periodicity)**.** *Suppose*

(a) $d_j = d_i$*;*

(b) $S_j(c+D) = S_i(c)$ *for every* $c \in V$*;*

(c) no square of A *is in* H_j*; and* $f(D) > 0$ *and* $\min_{c\in A} f(c) + f(D) > \max_{c\in H_j} f(c)$.

Then for every $k \geq 1$ *the motion from right turn* $i + kP$ *to right turn* $i + (k+1)P$ *is the motion from right turn* i *to right turn* j *moved by* kD*. In particular* $p_{i+kP} = p_i + kD$*, and the ant moves away to infinity along* D*.*

Proof. We prove by induction on $k \geq 1$ the statement

$$\mathcal{P}(k): \quad p_{i+kP} = p_i + kD, \qquad d_{i+kP} = d_i, \qquad S_{i+kP}(c+kD) = S_i(c) \text{ for all } c \in V,$$
$$H_{i+kP} = H_j \cup \textstyle\bigcup_{l=1}^{k-1}(V + lD).$$

$\mathcal{P}(1)$ is the definition of D together with (a) and (b).

Assume $\mathcal{P}(k)$. Between right turn i and right turn j the ant reads and writes only squares of V, starting from p_i, d_i and the colours S_i on V. By $\mathcal{P}(k)$, at right turn $i + kP$ the ant is at $p_i + kD$, faces d_i, and the colours on $V + kD$ are those of S_i on V, moved by kD. So the next P right turns are the motion from i to j moved by kD: the ant visits exactly $V + kD$, ends at $p_i + (k+1)D$ facing $d_j = d_i$, and $S_{i+(k+1)P}(c + kD) = S_j(c)$ for every $c \in V$. This gives the position, direction and visited set in $\mathcal{P}(k+1)$. For the colours, take $c \in V$ and put $z = c + (k+1)D$.

If $c + D \in V$, then $z = (c + D) + kD$, so $S_{i+(k+1)P}(z) = S_j(c + D) = S_i(c)$ by (b).

If $c + D \notin V$, then $c + D \in A$. We claim z has never been visited. First, $f(z) = f(c + D) + kf(D) \geq \min_A f + f(D) > \max_{H_j} f$, so $z \notin H_j$. Next, if $z \in V + lD$ with $1 \leq l \leq k$, then $c+(k+1-l)D \in V$. For $l = k$ this says $c+D \in V$, which is false. For $l < k$ it says $c+mD \in V$ with $m = k+1-l \geq 2$; but $f(c+mD) = f(c+D)+(m-1)f(D) \geq \min_A f+f(D) > \max_{H_j} f \geq \max_V f$, since $V \subseteq H_j$, so this is false too. Hence $z \notin H_{i+(k+1)P}$ and $S_{i+(k+1)P}(z) = 0$. On the other side, $S_i(c) = S_j(c + D)$ by (b), and $c + D \in A$ is not in H_j by (c), so $S_j(c + D) = 0$. Thus $S_{i+(k+1)P}(z) = S_i(c)$ in this case as well, which proves $\mathcal{P}(k+1)$.

Since $f(p_{i+kP}) = f(p_i) + kf(D)$ with $f(D) > 0$, the ant moves away to infinity. □

Remark 11. In the symbolic run, condition (b) compares affine expressions; equal expressions give equal colours for every K, so (b) holds for every $K \geq 4$ at once. Conditions (a) and (c) involve only positions and directions, which do not depend on K.

The program `proof/highway.py` takes $i = 7{,}600$ and $P = 10$. It finds $D = (-2, 2)$, a step count of $16K - 12$ for the period, and conditions (a), (b) (on the 18 squares of V) and (c) all true, with every decision valid for $K \geq 4$. Lemma 10 gives (iii) for $K \geq 4$.

6.5 A certificate that can be checked independently

So that the proof does not rest on trusting one program, the symbolic run also writes a *certificate* (`proof/certificate.txt`): a list of every action it takes up to its 7,610-th right turn, each with the numbers it depends on. There are two kinds of line: a single step (square, direction, the square's count $a + bK$ before the visit, and the turn `L` or `R`) and a spin (square, direction, and the number of laps $n = a + bK$). The certificate has 12,941 lines: 9,135 single steps, 3,805 spins and one mark where the ant leaves the box B.

A separate program, `proof/check_certificate.py`, which shares no code with the proof program, replays the ant from the empty grid line by line. It makes no choices; it only checks. Every check has the form "$a + bK \geq c$ for every $K \geq 4$", which holds exactly when $b \geq 0$ and $a + 4b \geq c$. The checker verifies that:

- each line's square, direction and count agree with its own replay;
- every count stays a colour, $0 \leq a + bK \leq K - 1$, for every $K \geq 4$;
- a single step turns `R` exactly when the count is $-1 + 1K = K - 1$, and turns `L` only when the count is at most $K - 2$ for every $K \geq 4$;

- a spin satisfies Lemma 9: $n \geq 1$, and $n \leq (K-1) - \sigma(c_i)$ for each of its four squares, for every $K \geq 4$; before the edge is reached, no spin touches the edge of B;
- the edge mark is exactly where the replay first leaves B.

In total it checks 69,260 inequalities, each valid for every $K \geq 4$. From its own replay it then finds the edge after 7,488 right turns at step $11{,}980K - 8{,}971$, and the three conditions of Lemma 10 at right turns 7,600 and 7,610, with shift $(-2, 2)$ and $16K - 12$ steps per period. As a test of the checker, we altered the certificate in several ways (one spin made one lap longer, one left turn changed to a right turn, one line deleted); each change was rejected at the line where it was made.

6.6 The case $K = 3$

For $K = 3$ the rule is `LLR` and everything is an ordinary number. A direct run (`src/ant_visits.c`) reaches the edge at step 26,969 after 7,488 right turns (`proof/skeleton.py` checks the right turns square by square against $K = 4, 5, 6, 7, 10, 20$), and `proof/highway_concrete.py` verifies conditions (a)–(c) of Lemma 10 with $i = 7{,}600$, $P = 10$, $D = (-2, 2)$ and a period of $36 = 16{\cdot}3 - 12$ steps. This completes the proof of Theorem 8. □

7 Discussion

What the proof shows. For $\mathtt{L}^{K-1}\mathtt{R}$ the number of colours only changes *how long* the ant spins, never *where* it turns right. A single symbolic run therefore covers infinitely many ants. The same method applies to any family of rules in which a parameter changes only the length of spins, and could be tried on other families found in the census, for example $\mathtt{L}^a\mathtt{R}^b$ patterns.

Limits of the census. The classes describe each picture after 10^6 steps. A machine counted as chaotic may still build a highway later: the slowest highway found took 937,513 steps to reach the edge, and for the classic ant it is not even proved that a highway must come from every finite start. Two machines checked for 2×10^9 steps, `LLLRL` (a blob 1,954 squares wide after 1.024×10^9 steps) and `LLLRR` (a disc 49 squares wide), built no highway in that time. The tests are rough: a few blobs sit inside square frames, and one wedge-shaped machine, `LLRLRRLLL`, grows a chequered arrowhead that has no class of its own and is counted as chaotic. All runs start from an empty grid.

Further work. Natural next steps are (1) running the chaotic machines much longer to find late highways; (2) a census with the copies of Section 4.1 removed for every class, not only highways; (3) proving the even-runs pattern and the slow "disc" growth; and (4) submitting the count sequences by K to the OEIS once they are stable.

Code and data availability

All code and data, including the proof certificate and its checker, are in the repository

`https://github.com/VishnuVaidyanathan/langtons-ant-census`

The script `run_all.sh` rebuilds every number, table and figure in this paper in about two minutes on a laptop. Appendix A lists the files.

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A Repository contents

File	Purpose
`run_all.sh`	Rebuilds everything from scratch.
`src/ant_census.c`	Runs all 2^K rule strings for one K; prints shape measures.
`src/classify.py`	Removes repeated strings and mirror twins; assigns classes.
`src/analyse.py`	All numbers of Sections 3–5.
`src/ant_visits.c`	Edge step and most visits to one square for one rule.
`src/ant_snap.c`, `src/make_figures.py`	Figures.
`proof/symbolic.py`	Symbolic run: Theorem 8(i)–(ii) for $K \geq 4$.
`proof/highway.py`	Symbolic periodicity check: Theorem 8(iii) for $K \geq 4$.
`proof/write_certificate.py`	Writes the certificate `proof/certificate.txt`.
`proof/check_certificate.py`	Independent checker of the certificate (Section 6.5).
`proof/highway_concrete.py`	The same check with plain numbers, used for $K = 3$.
`proof/skeleton.py`	The observations behind the proof, for several K.
`data/census.csv`	The 4,015 distinct machines with class and measures.
`data/census_raw.csv`	Measures for all 8,188 rule strings.
`data/highways.txt`, `data/squares.txt`	Highway and square machines by K.